• Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH (BIOTECHNOLOGY)

END SEMESTER EXAMINATIONS, NOVEMBER 2024 (REGULAR)

SUBJECT: BIOREACTION ENGINEERING (BIO 3104)

REVISED CREDIT SYSTEM

Time: 3 Hours (26/11/2024) MAX. MARKS: 50

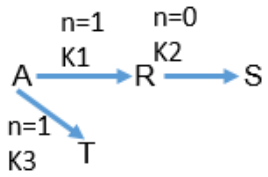
Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitable assumed.

SL No		Marks
1A.	Explain how to find given reaction is elementary or non-elementary.	**2**
1B.	The secondary metabolites (P and Q) are produced using *Bacillus* species in submerged fermentation process. The following mechanism has been proposed. A $\Leftrightarrow$ P + Q* Q* $\rightarrow$ R* +S R*+Q*$\rightarrow$ 2P A is a substrate, P and S are metabolic products and R*& Q* are intermediates. Prove that above proposed mechanism is consistent with and can explain the	**4**

<table>
<tr><td></td><td>observed first order decomposition of substrate A.</td><td></td></tr>
<tr><td>1C.</td><td>With the suitable example explain the parameters affecting the rate of reaction in heterogeneous reaction system. Write various definitions reaction rate in heterogeneous reaction system</td><td>4</td></tr>
<tr><td>2A.</td><td>What do you mean by shifting order reactions? How do find the kinetics in the following shifting order reactions
Shift from low to high order as the concentration drops
Shift from high to low order as the concentration drops</td><td>3</td></tr>
<tr><td>2B.</td><td>)llowing data are obtained at 0ºC in a constant-volume batch reactor
pure gaseous A. The stoichiometry of the decomposition is A → R.
rate equation which satisfactorily represents this decomposition
<table><tr><td>Time, min</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td><td>∞</td></tr><tr><td>Partial Pressure of A, mm</td><td>760</td><td>600</td><td>475</td><td>390</td><td>320</td><td>275</td><td>240</td><td>215</td><td>150</td></tr></table></td><td>4</td></tr>
<tr><td>2C</td><td>What do you mean by homogeneous catalyzed reactions? How do you plan your kinetic runs to evaluate the kinetics of homogeneous catalyzed reaction of the following elementary scheme:
A → R
A +C → R+C</td><td>3</td></tr>
</table>

3A.	The gas reaction 2 A → R + 2 S is approximately second order with respect to A, then pure A is introduced at 1 atm into a constant volume batch reactor, the pressure rises 40 % in 3 min. For a constant pressure batch reactor find The time required for the same conversion b. The fractional increase in volume at that time	**3**
3B.	Hydrolysis of sucrose is carried out in an immobilized invertase enzyme packed bed bioreactor system in series that behaves like a PFR. The multiple reactor system consists of a number of Perspex columns with each 5cm diameter and Length=75 cm in series. The substrate sucrose (S_0=4M) is pumped at 0.1 L/min at entrance of the reactor system. The enzyme catalyzed reaction follows the substrate inhibition kinetics. Find the volume of the reactor to achieve 60% conversion at steady state. Also find the number of tubes required to achieve this conversion with above specifications. Substrate inhibition kinetics: $-rs = \frac{Vm.S}{(Km + S + \frac{S^2}{KI})}$ moles/liter.min Kinetics data: V_m=0.028 M/min, K_m=0.23 M, K_I=0.2 M	**4**
3C	The kinetics of the aqueous-phase decomposition of A is investigated in two mixed flow reactors in series, the second having twice the volume of the first reactor. At steady state with a feed concentration of 1 mol A/liter and mean residence time of 96 sec in the first reactor, the concentration in the first reactor is 0.5 mol A/liter and in the second is 0.25 mol A/liter. Find the kinetic equation for the decomposition.	**3**

<table>
<tr>
<td>4A.</td>
<td>The kinetic study was carried out in a batch reactor in the presence of Enzyme and shown in the following table. Graphically Design a best reactor system of two MFR’s in series to obtain 90 % conversion with CA0=1 M, v0=10 L/min
<table>
<tr><td>CA, M</td><td>0.05</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td><td>0.6</td><td>0.7</td><td>0.9</td><td>1.2</td></tr>
<tr><td>-rA, M/min</td><td>0.022</td><td>0.033</td><td>0.067</td><td>0.10</td><td>0.133</td><td>0.132</td><td>0.099</td><td>0.067</td><td>0.04</td><td>0.029</td></tr>
</table>
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<td>4</td>
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<tr>
<td>4B.</td>
<td>For the following First order followed by zero order series reaction that is taking place in a batch fermenter


Find the $C_{R,\,max}$ and t_{max}</td>
<td>3</td>
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<tr>
<td>4C</td>
<td>What are the advantages of the Recycle reactor? Write the expression for optimum recycle ratio and represent it graphically for the typical elementary autocatalytic reaction of A + R →R + R</td>
<td>3</td>
</tr>
<tr>
<td>5A.</td>
<td>Chemostat of 1000 liters capacity is used for the production biomass using glucose substrate. The microbial system follows a Monod growth kinetics with μ_m =
, K_S = 1.5 g/l and the yield factor $Y_{X/S}$ = 0.5 gbiomass/g substrate consumed.
erile feed containing 10g/l substrate is pumped at 100 L/h to the chemostat.
Determine the substrate and biomass concentrations at steady state
Estimate the washout dilution rate</td>
<td>4</td>
</tr>
</table>

5B.	Explan in the various accessories needed for the operation of batch fermenter with ceultures. Derive the expression for the batch reaction time for the conversion of substrate from S_0 to S_f during batch cultivation of bakers yeast	**3**
5C	Aqueous biochemical reaction is carried out in a plug flow reactor with v_0=4 liter/min, C_{A0}=200 mol/liter, C_{R0}=20 mol/liter. This reaction is reversible and represented by A $\Leftrightarrow$ R ,$-r_A$= (0.05 min^{-1}) C_A - (0.02 min^{-1}) C_R. Find the volume of PFR required to achieve 80 % of maximum possible conversion.	**3**